Wanyue (Jenny) Lin

Berkeley, CA | jennylinwanyue@eecs.berkeley.edu | (314) 435-2785
Google Scholar

EDUCATION

University of California, Berkeley Aug 2025 – Present
Ph.D. in Electrical Engineering and Computer Sciences (EECS); expected graduation: 2030 or later.
Research interests: Stochastic Control, Formal Methods, Temporal Logic, Stochastic Optimization, Safe and Reliable Autonomous Systems.

Washington University in St. Louis Aug 2022 – May 2025
B.S. in Systems Science and Engineering; second major in Physics; minors in Electrical Engineering and Music.
GPA: 3.91/4.00.

PUBLICATIONS AND MANUSCRIPTS

Manuscript submitted for review

"Data-Driven Verification and Parameter Synthesis for Stochastic Signal Temporal Logic Specifications." *Submitted to Design, Automation & Test in Europe (DATE 2027).*

Peer-reviewed publications

Y. Wu, Y. Liu, **W. Lin**, Z. Nussinov, and S. Ran, "Machine-learning prediction of superconductivity from high-temperature resistivity in Fe-based compounds," *APS Open Science*, vol. 1, 000053, 2026.

W. Lin and A. Clark, "Optimal Beamforming Design for Stability and Performance of UAV Formations," *American Control Conference (ACC)*, pp. 476–483, 2025.

C. Broyles, X. Wan, W. Cheng, et al. (including **W. Lin**), "High-temperature surface state in Kondo insulator $U_3Bi_4Ni_3$," *Science Advances*, vol. 11, no. 12, eadq9952, 2025.

W. Lin, Y. Wu, C. Broyles, T. Kong, and S. Ran, "A single crystal study of Kagome metals U_2Mn_3Ge and U_2Fe_3Ge," *Journal of Physics: Condensed Matter*, vol. 36, no. 34, 345602, 2024.

RESEARCH EXPERIENCE

University of California, Berkeley | Doctoral Researcher Aug 2025 – Present

Project 1: Control Synthesis for Stochastic Systems under StSTL Specifications Ongoing

- Developing a control-synthesis framework for discrete-time stochastic linear systems subject to polyhedral stochastic signal temporal logic (StSTL) specifications and uncertain disturbances.
- Formulating finite-horizon optimization of control inputs and StSTL satisfaction robustness; investigating nominal-control/feedback decomposition with data-driven probabilistic constraint tightening and feedback-gain design.
- Studying finite-sample, probably approximately correct (PAC) certification for nested logical and temporal specifications, with attention to theoretical assumptions and computational tractability.

Project 2: Data-Driven StSTL Verification and Parameter Synthesis DATE 2027 submission

- Contributed to a distribution-free framework for quantitative verification and parameter synthesis of bounded-time StSTL specifications from sampled stochastic trajectories.
- Applied one-sided Clopper–Pearson confidence bounds and sample-and-discard scenario optimization to derive finite-sample probabilistic guarantees for atomic predicates, then propagate robustness bounds through Boolean and temporal operators.
- Studied monotone parameter synthesis using one-dimensional scenario subproblems and mixed-integer linear programming; the submission includes adaptive-cruise-control and drone-flight examples.

ADDITIONAL RESEARCH EXPERIENCE

CyTrus Lab, Washington University in St. Louis | Andrew Clark Sep 2023 – May 2025
Networked Control, Wireless Communication, and Cyber-Physical Systems

- Developed stochastic wireless communication and UAV-formation control models under packet loss and channel fading; derived Lyapunov-based stability conditions for networked control.
- Formulated joint communication–control optimization balancing state-estimation error, communication cost, power constraints, and closed-loop stability.
- Used Bernstein-type probability bounds and semidefinite programming to optimize beamforming weights and transmit power; validated performance using MATLAB simulations (*ACC 2025*).
- Completed an honors thesis on control–communication co-design for multi-agent cyber-physical systems, including value-of-information-aware wireless resource allocation.

Ran Lab, Washington University in St. Louis | Sheng Ran Dec 2022 – Sep 2025
Condensed Matter Physics and AI for Science

- Led single-crystal growth and experimental characterization of Kagome compounds U_2Mn_3Ge and U_2Fe_3Ge; analyzed electrical transport, magnetization, and heat-capacity measurements (*J. Phys.: Condens. Matter*, 2024).
- Contributed to research on surface transport in the Kondo insulator $U_3Bi_4Ni_3$ (*Science Advances*, 2025).
- Co-authored a machine-learning study using high-temperature resistivity measurements (150–300 K) to predict superconductivity in Fe-based compounds (*APS Open Science*, 2026); also explored XGBoost and CNN models for related resistivity prediction tasks.

Lu Lab, Washington University in St. Louis | Chenyang Lu Mar 2024 – May 2025
Machine Learning for Clinical Decision Support

- Built predictive models for hospital readmission using a dataset of approximately 7,000 cancer patients; performed preprocessing, feature selection, and model evaluation with logistic regression, XGBoost, and variational autoencoders.
- Collaborated with clinical and data-engineering partners on model integration with the Epic system; explored large-language-model methods for synthetic tabular data.

SELECTED RESEARCH PRESENTATIONS

- **Optimal Beamforming Design for Stability and Performance of UAV Formations**: honors thesis podium presentation (Dec 2024); undergraduate research poster (Nov 2024); research lightning talk (Jul 2024).
- **Kagome Metals U_2Mn_3Ge and U_2Fe_3Ge**: undergraduate research poster (Apr 2024); physics research symposium oral presentation (Nov 2023).

TEACHING AND MENTORING

- **Teaching Assistant / Grader**, ESE 415 Optimization and ESE 417 Machine Learning, Washington University in St. Louis (Sep 2024 – May 2025): office hours, grading, and instructional support.
- **Engineering Tutor**, General Physics and Engineering Mathematics (Sep 2023 – May 2024); **Peer Learning Leader**, Multivariable Calculus and Introduction to Engineering (2023).

TECHNICAL SKILLS

Programming and Tools: Python, MATLAB, C++, Java, Mathematica, LaTeX, Linux, Origin.
Methods: Stochastic and convex optimization, semidefinite programming, model predictive control, formal verification, probabilistic modeling, machine learning.
Research and Engineering: Experimental data analysis, SolidWorks, CNC machining, laser cutting, 3D printing, soldering.